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MULTIPLIER OPTIMIZATION FOR CONSTANT PROPORTION PORTFOLIO INSURANCE (CPPI) STRATEGY

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Constant proportion portfolio insurance (CPPI) strategy is a very popular investment solution which provides an investor with a capital protection as well as allows for an equity market participation. In this paper, we propose a two-step approach to the numerical optimization of the CPPI main parameter, multiplier. First, we identify an admissible range of the multiplier values by controlling the shortfall probability (chosen as a measure of the gap risk). Second, within the admissible range, we choose the optimal multiplier value with respect to the omega ratio (chosen as a performance measure). We illustrate the performance of our optimization algorithm on simulated CPPI paths in the Black–Scholes environment with discrete trading as well as on the historical S&P500 data using the block-bootstrap simulations.

Keywords: CPPI strategy; multiplier; constrained optimization; gap risk; bootstrap simulations.

1. Introduction

In the current capital market environment, investors constantly face the challenge of finding a successful and stable investment mechanism. Highly volatile equity markets and extremely low bond returns bring about the demand for sophisticated and reliable risk management strategies. Therefore, investors are looking for risk management solutions to effectively protect their investments. One of the most prominent strategies commonly used in the financial industry is constant proportion portfolio insurance (CPPI) strategy.

The CPPI is a dynamic asset management approach that provides a certain level of capital protection of the investment (a guarantee of a certain investment value at maturity). Introduced by Perold & Sharpe (1988) for fixed-income instruments and

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by Black & Jones (1987) for equity instruments, it efficiently protects an investment from unpredictable market downturns and allows for significant participation in a rising market by dynamically reallocating the investment between risky and risk-free assets. The CPPI method has gained popularity due to its simplicity and transparent algorithm. It also implies no fee (despite transaction costs) and can be terminated at any time since it is not tied to any financial derivatives.

The past research has been focused on analyzing the performance of the CPPI strategy with respect to other strategies rather than as an optimization process for the CPPI strategy alone. The CPPI strategy is often compared to the option based portfolio insurance (OBPI) strategy (e.g. Bertrand & Prigent 2005, Zagst & Kraus 2011, Maalej & Prigent 2016, Pézier & Scheller 2013) with results indicating that the CPPI strategy is almost never inferior to the OBPI one. If the volatility spread for the option pricing in the OBPI is taken into account, the CPPI superiority becomes even more pronounced. Dichtl *et al.* (2017) compare the CPPI, TIPP ("refined" CPPI), stop-loss, synthetic put, and dynamic value at risk strategies and find that CPPI and TIPP offer the best downside protection.

Since the CPPI strategy often takes a leading place among capital protection mechanisms, it remains a widely used investment solution. Therefore, the question of the optimal CPPI design arises: what parameters and rebalancing rules to choose for the CPPI strategy set-up? The simple CPPI investment is characterized by two parameters: a protection level P and a multiplier m. The goal of this paper is to suggest a mechanism for the CPPI optimization with respect to its multiplier m. It is a crucial part of the strategy set-up. If a multiplier is set lower than necessary, an investor misses a chance to benefit from a rising market. If a multiplier is set too high, the product provider may suffer significant losses compensating the client for missing guaranteed amount.

The current body of literature provides mixed suggestions on the optimal multiplier value. For example, Bertrand & Prigent (2005) and Zagst & Kraus (2011) choose the multiplier based on the stochastic dominance of the CPPI strategy over the OBPI one. They look at very conservative multipliers ($m < 5$). Herold *et al.* (2007) use a multiplier $m = 5$ for a risky asset referencing it as a typical multiplier for stock investments in practice. Such a multiplier value is also often used for portfolio simulations when comparing the CPPI strategy to other portfolio insurance strategies (e.g. Dichtl *et al.* 2017). It carries an implicit assumption that losses up to 20% overnight can occur before an investor has a chance to rebalance their portfolio (assuming daily rebalancing). However, there is no evidence that this is a realistic expectation of the market dynamics. For instance, even during the global financial crisis in years 2008–2009, the sharpest overnight drop in the price of S&P500 Index was 9.03%,[a] which allows a multiplier as high as $m = 11$ without violating the protection level of the CPPI strategy. Dichtl & Drobetz (2010) argue that under the prospect theory, investors should choose the highest possible multiplier constrained

[a]However, this number only reflects the biggest overnight drop in closing prices of the S&P500 Index.

by the maximum acceptable risk of the violated protection of the investment. Other works fix the multiplier in such a way that leads to a specific initial exposure to the risky asset. For example, Annaert *et al.* (2009) evaluate the CPPI portfolio with a multiplier $m = 14$ and protection level $P = 95\%$ because they set the exposure to equity at 70%, Ho *et al.* (2011) analyze the strategy with a multiplier around three to five to keep the initial exposure at 25%, and Cesari & Cremonini (2003) simulate a CPPI portfolio with m set in such way that the exposure equals 50%. Zhang *et al.* (2015) run the CPPI strategy with a multiplier $m = 3$. It becomes clear that there is no consensus on the optimal or appropriate value of the multiplier as well as no comprehensive framework to select one even given a constraint pool of options. An exception is a dynamic choice of the multiplier proposed by Hamidi *et al.* (2014), which, however, offers a strategy that goes beyond a simple CPPI method.

We, therefore, propose a straightforward and clear constrained optimization method for a simple CPPI strategy that consists of two steps: first, we limit the possible choice of the multipliers by controlling the gap risk under a given threshold, and second, we choose the best multiplier based on the portfolio performance measure, omega ratio. We show how this optimization method works with two different approaches to portfolio simulations: a Monte Carlo simulation approach under the assumption of the Black–Scholes market with discrete rebalancing and a block-bootstrap simulation approach for which we use historical data on the daily closing prices of the S&P500 Index.

This paper is organized as follows: following this introduction, we explain in Sec. 2 how the CPPI strategy works. We separately discuss the continuous trading case and the discrete trading case. In Sec. 3, we present our optimization approach. We first introduce the criteria for the optimal multiplier choice (Secs. 3.1 and 3.2). We formulate our constrained optimization problem and outline our proposed numerical optimization algorithm in Sec. 3.3. Then the numerical optimization results for two different simulation methods are presented in Secs. 3.4 and 3.5. In Sec. 4, we use historical market data to illustrate the performance of our optimization algorithm in different market scenarios. Finally, Sec. 5 concludes our discussion.

2. The CPPI Strategy

Let us recall the construction of the CPPI strategy. In its basic version, the CPPI strategy is determined by three main constituents: a floor, a cushion, and a multiplier. A multiplier is a constant which is set-up at time zero and is kept unchanged until the investment maturity. The floor and the cushion are dynamically determined at every portfolio rebalancing time.

Let V_t^{CPPI} be the value of the CPPI portfolio at time t, P be a protection level of the strategy, T be a contract expiration time, and r be a risk-free rate. Then at time t, the floor value F_t is calculated as follows:

$$F_t = e^{-r(T-t)} P V_0^{\mathrm{CPPI}}. \tag{2.1}$$

Here, V_0^{CPPI} is the initial investment in the CPPI strategy.

A cushion C_t at time t is the surplus of the portfolio value V_t^{CPPI} over the floor F_t:

$$C_t = V_t^{\mathrm{CPPI}} - F_t. \tag{2.2}$$

Finally, a multiplier m is a prespecified parameter that determines the riskiness of the strategy. The multiplier is used to determine the exposure, i.e. the amount to be invested in a risky asset. At time t, the exposure E_t is defined as follows:

$$E_t = mC_t, \tag{2.3}$$

where C_t is defined in (2.2). The remaining part B_t of the portfolio value at time t is invested in a risk-free asset:

$$B_t = V_t^{\mathrm{CPPI}} - E_t. \tag{2.4}$$

At every rebalancing time, the values of the floor and the cushion are reassessed and the investment is redistributed between the risky and risk-free assets.

The terminal value of the floor is the guaranteed value of the investment at maturity:

$$F_T = PV_0^{\mathrm{CPPI}}. \tag{2.5}$$

In the current market environment, it is hardly possible to provide a full guarantee at the end of the contract (100% protection, or $P = 1$) due to extremely low interest rates. Setting the level of guarantee P too high prevents an investor from having the exposure to the market. Therefore, the industry often offers protection levels between 80% and 90%. Moreover, if the insured percentage rises, the expected return of the CPPI strategy decreases (Bertrand & Prigent 2005).

In a simple set-up of the CPPI strategy, the multiplier m is constant throughout the whole lifetime of the CPPI portfolio. In a bearish market, the higher multiplier would result in a portfolio which approaches the floor faster. In contrast, in a bullish market, the CPPI strategy with a high multiplier would result in higher portfolio returns.

Insurances with a multiplier conditional on market volatility (e.g. Lee *et al.* 2008) or on other risk measures such as value at risk (e.g. Hamidi *et al.* 2014) belong to the class of variable proportion portfolio insurances (VPPI) or dynamic proportion portfolio insurances (DPPI). Another modification of the CPPI strategy, time-invariant portfolio protection (TIPP), consists in locking up an increase in portfolio value by calculating the floor value as guarantee P from the highest value of the portfolio since the beginning of the contract. However, research shows that TIPP strategy provides no significant improvement relative to the classic CPPI strategy (Dichtl *et al.* 2017).

The main concern of the investors when considering the CPPI portfolio is that there is a chance, before the contract maturity, that the portfolio is completely invested in cash (a so-called cash-lock case) and, therefore, it cannot gain any profit from the rising market. Such a scenario takes place when an investment faces a falling market and reaches the floor shortly after it was launched. If this happens, the investor has no exposure to the market when later an underlying risky asset goes up in price. In addition, this strategy is vulnerable to a falsely estimated expected maximum risk: the floor can be breached if a loss between two consecutive rebalancing times exceeds the assumptions reflected in the multiplier. However, the breach will happen only if there is a very sharp drop in the market before the investor has a chance to trade.[b]

2.1. *Continuous time rebalancing in the Black–Scholes market*

It is often assumed in academic studies related to the CPPI strategy that (i) the underlying risky asset follows the geometric Brownian motion (GBM) and (ii) there are no trading restrictions, meaning that trading is continuous.

Under the first assumption, the dynamics of a risky asset price S_t in time is defined by a standard Brownian motion W_t, drift $\mu > r > 0$, and volatility σ as follows:

$$dS_t = S_t(\mu dt + \sigma dW_t). \tag{2.6}$$

The growth of a risk-free asset B_t is determined by a constant continuous interest rate $r > 0$:

$$B_t = B_0 e^{rt}. \tag{2.7}$$

Under the idealistic assumptions that the financial market allows for continuous trading and follows the Black–Scholes model, the closed form expression for the value of the CPPI portfolio at time t can be found by means of the Itô lemma (see Bertrand & Prigent 2005, for the proof):

$$V_t^{\text{CPPI}} = F_t + C_t = e^{-r(T-t)} P V_0^{\text{CPPI}} + C_0 \left(\frac{S_t}{S_0}\right)^m e^{(1-m)(r+\frac{1}{2}m\sigma^2)t}. \tag{2.8}$$

Here, P and m are the parameters of the CPPI strategy, the protection level and the multiplier, respectively; V_0^{CPPI} is an initial investment in the strategy, F_t is a floor value at time t [see (2.1)], C_t is a cushion value at time t [see (2.2)]. We remark here that it follows from (2.1) and (2.2) that the initial value of the cushion C_0 is given by the following formula:

$$C_0 = V_0^{\text{CPPI}}(1 - Pe^{-rT}). \tag{2.9}$$

Further, the expected value and the variance of the CPPI portfolio value at the terminal time T of the investment period are given, respectively, by:

$$E[V_T^{\text{CPPI}}] = PV_0^{\text{CPPI}} + C_0 e^{[r+m(\mu-r)]T}, \tag{2.10}$$

[b]In practice, investors face additional risks related to market liquidity, as well as to high transaction costs. We leave these factors outside of the current investigation.

and

$$\mathrm{Var}[V_T^{\mathrm{CPPI}}] = C_0^2 e^{2[r+m(\mu-r)]T}(e^{m^2\sigma^2 T} - 1), \tag{2.11}$$

(see Zagst & Kraus 2011, for details).

It follows from (2.10), that the expected value of the terminal CPPI investment value is independent of the volatility σ. This makes an intuitive sense in a market where trading can happen instantaneously, but seems unrealistic in the case of a discrete trading.

The closed form formulas (2.8)–(2.11) allow one to make straightforward estimates on how the CPPI parameters affect the terminal value of the CPPI portfolio. The expected value of the CPPI portfolio increases with an increase in the multiplier m. The CPPI variance increases exponentially with an increase in volatility σ. This growth is further amplified by the higher values of the multiplier m. The higher protection level P leads to the lower expected value as well as the lower variance of the terminal CPPI portfolio value.

The portfolio will approach the floor in the bearish market faster with higher values of the multiplier m. However, the Black–Scholes model settings and continuous trading ensure that the guarantee is never violated and the CPPI portfolio V_t^{CPPI} is always above the current floor F_t, the discounted value of the guaranteed amount. This is because the cushion C_t is always above zero. Thus, there is no upper limit on the multiplier m. However, the higher the multiplier m, the faster the portfolio will approach the floor in the bearish market.

2.2. *CPPI with trading restrictions. Gap risk*

Continuous time trading is an unrealistic assumption in a real financial market. Normally, the trading is restricted to a discrete case. It is therefore important to evaluate a portfolio insurance strategy under discrete trading. Let us consider a CPPI portfolio with the lifetime period $[0, T]$. The portfolio is rebalanced at equidistant discrete times $t_1, t_2, \ldots, t_n = T$. The value of the CPPI portfolio at any given rebalancing time t_k, $(k = 1, \ldots, n)$ is

$$V_{t_k}^{\mathrm{CPPI}} = (V_{t_{k-1}}^{\mathrm{CPPI}} - mC_{t_{k-1}})e^{r\frac{T}{n}} + \frac{mC_{t_{k-1}}}{S_{t_{k-1}}} S_{t_k}. \tag{2.12}$$

The CPPI strategy in the market with discrete trading is studied, for example, in Balder *et al.* (2009) and Bertrand & Prigent (2002). The authors lift the assumption of a continuous-time trading, but keep the Black–Scholes environment. It allows them to derive a closed-form formula for the expected terminal value of the CPPI portfolio:

$$E[V_T^{\mathrm{CPPI}}] = P\,V_0^{\mathrm{CPPI}} + (V_0^{\mathrm{CPPI}} - F_0)\left[E_1^n + e^{-r\frac{T}{n}} E_2 \frac{e^{rT} - E_1^n}{1 - E_1 e^{-r\frac{T}{n}}}\right], \tag{2.13}$$

where

$$E_1 = me^{\mu\frac{T}{n}}\mathcal{N}(d_1) - e^{r\frac{T}{n}}(m-1)\mathcal{N}(d_2), \tag{2.14}$$

$$E_2 = e^{r\frac{T}{n}}[1 + m(e^{(\mu-r)\frac{T}{n}} - 1)] - E_1. \tag{2.15}$$

Here, d_1 and d_2 are defined as

$$d_2 = \frac{\left(\ln\dfrac{m}{m-1} + (\mu - r)\dfrac{T}{n} - \dfrac{1}{2}\sigma^2\dfrac{T}{n}\right)}{\left(\sigma\sqrt{\dfrac{T}{n}}\right)} \quad \text{and} \quad d_1 = d_2 + \sigma\sqrt{\frac{T}{n}}. \tag{2.16}$$

It follows from (2.13) (see Balder *et al.* 2009) that the expected terminal value of the CPPI portfolio increases when a multiplier m and the asset expected rate of return μ increase, and decreases with an increase in the protection level P. In contrast with the continuous trading case, the higher market volatility σ also leads to the higher expected CPPI return.

In some CPPI applications, the tolerance is included as an additional component of the CPPI strategy (Black & Jones 1987). The tolerance defines the percentage in the risky asset price movement that would trigger a trade. For example, consider a CPPI strategy where the rebalancing happens only if the market moves by at least 2%. However, this component of the strategy is often omitted in the discussions of the CPPI strategy. Particularly, the strategy is evaluated based on the rebalancing at constant intervals such as daily, weekly, monthly, quarterly, and yearly. Using Chinese market data, Zhang *et al.* (2015) show that the trigger-induced rebalancing and the rebalancing with equal time intervals result in no significant difference in the CPPI performance. Annaert *et al.* (2009) argue that the daily rebalancing is the preferred frequency even when transaction costs are taken into consideration.

In the case of a discrete-time trading, there is a positive probability that a CPPI portfolio will fall below the floor before an investor has a chance to rebalance their portfolio. Such situation may occur if the risky asset price drops significantly between the rebalancing times. This results in an unavoidable risk that the terminal value of the investment will not meet the guaranteed protection requirement. The risk of violating the guarantee level of the investment is called *a gap risk*. In the case of a continuous trading, the CPPI strategy experiences no gap risk: an investor has a chance to trade instantly as the market experiences sharp drops. In practice, trading is restricted to discrete times (e.g. daily).

It is straightforward to show (see e.g. Bertrand & Prigent 2002) that the following condition on the multiplier m theoretically eliminates a gap risk in the case of the discrete rebalancing with n rebalancing times over the time interval $[0, T]$: for any rebalancing time t_k $(k = 1, \dots, n)$

$$m\frac{S_{t_k}}{S_{t_{k-1}}} - (m-1)\, e^{r\frac{T}{n}} \geq 0 \Leftrightarrow \frac{S_{t_k}}{S_{t_{k-1}}} \geq \frac{m-1}{m} e^{r\frac{T}{n}}. \tag{2.17}$$

If (2.17) holds for any rebalancing time t_k $(k = 1, \dots, n)$, the cushion C_{t_k} never falls below zero [see (2.2)]. If for some $k = 1, \dots, n-1$ (2.17) does not hold, the strategy fails to provide the guaranteed protection (see Bertrand & Prigent 2002, for more details).

Remark 2.1. Practitioners widely use a simplified version of the condition (2.17) based on the approximation $e^{r\frac{T}{n}} \approx 1$ for large n:

$$\frac{1}{m} \geq \frac{S_{t_{k-1}} - S_{t_k}}{S_{t_{k-1}}}, \tag{2.18}$$

for $S_{t_k} \leq S_{t_{k-1}}$. Inequality (2.18) suggests that $\frac{1}{m}$ should be greater than a percentage drop in the risky asset price over the time interval $[t_{k-1}, t_k]$, $(k = 1, \dots, n)$ between any two rebalancing points. Obviously, it is impossible to guarantee at time zero that (2.18) holds with probability one for any future rebalancing time t_k.

Analyzing the above inequalities, it is clear that the gap risk increases with an increase in a multiplier value. It is, therefore, necessary to address the risk that the strategy guarantee is breached.

3. Optimization Process: Methodology and Results

In this section, we suggest a two-step optimization algorithm for the CPPI strategy with respect to the multiplier m. We first outline the optimization criteria. Then we describe the algorithm implementation and present our numerical results.

3.1. *The measure of a gap risk*

The first step of our optimization algorithm consists of determining the admissible range of values for the CPPI multiplier by setting an upper limit on the gap risk.

We follow the work of Balder *et al.* (2009) in quantifying the gap risk as a shortfall probability P^{SF}, the probability that a portfolio does not meet its guarantee at maturity:

$$P^{\mathrm{SF}} = P(V_T^{\mathrm{CPPI}} \leq P V_0^{\mathrm{CPPI}}). \tag{3.1}$$

In order to maintain the initial goal of the CPPI strategy that is providing investment protection, the shortfall probability should be limited. This imposes an upper bound on the range of admissible multipliers for the strategy. As an implication of this constraint, we allow a multiplier to take only those values for which the probability of breaching the guarantee is sufficiently low (in this paper, we limit the shortfall probability by 1%).

Remark 3.1. Under the Black–Scholes model assumptions, Balder *et al.* (2009) derive a closed-form solution for the shortfall probability P^{SF} of a CPPI portfolio in a discrete trading environment:

$$P^{\mathrm{SF}} = 1 - (1 - \mathcal{N}(-d_2))^n, \tag{3.2}$$

for which n is the number of times portfolio is rebalanced (e.g. $n = 252$ if for daily rebalancing) and d_2 is defined in (2.16).

It is shown in Balder *et al.* (2009) that the shortfall probability increases with an increase in the multiplier m and in the market volatility σ. Therefore, in order to maintain an acceptable level of gap risk, an investor needs to impose an upper limit on the multiplier. Moreover, the higher market volatility, the stronger the restriction on the multiplier.

3.2. *The performance measure*

Once the admissible range of multipliers is determined at the first step of the optimization algorithm, the second step consists of choosing a multiplier for which the CPPI return distribution is "the best". There has been a great number of measures developed that compare the portfolio return distributions. We use the omega ratio (Keating & Shadwick 2002) as a comprehensive performance measure for portfolio insurances, proposed by Bertrand & Prigent (2011).

The probabilistic definition of the omega ratio is as follows. Let X be a random variable which represents a risky asset return over a certain fixed period of time. Let F be the cumulative distribution function of X. Let (a, b) be a support of the probability distribution for X. Denote by L a prespecified loss threshold. Then the omega ratio of X with the threshold L is defined as follows:

$$\Omega_X(L) = \frac{\displaystyle\int_L^b (1 - F(x))dx}{\displaystyle\int_a^L F(x)dx}. \tag{3.3}$$

It is shown in e.g. Kazemi *et al.* (2004) that (3.3) is equivalent to:

$$\Omega_X(L) = \frac{E[\max(X - L, 0)]}{E[\max(L - X, 0)]}, \tag{3.4}$$

where both expectations are taken with respect to the probability distribution of X. So, the omega measure is calculated as a ratio of the expected returns above the given threshold to the expected returns below this threshold.

3.3. *Optimization problem and algorithm implementation*

Let X be the percentage return of the CPPI portfolio over the period $[0, T]$:

$$X = \frac{V_T^{\mathrm{CPPI}} - V_0^{\mathrm{CPPI}}}{V_0^{\mathrm{CPPI}}}. \tag{3.5}$$

Let L be a prespecified threshold and let p be a prespecified upper limit on the shortfall probability. Summarizing the discussion in the previous two subsections, let us formulate the constrained optimization problem that we solve numerically in what follows:

$$\text{Maximize } \ \Omega_{X,m}(L) \tag{3.6}$$

$$\text{subject to } \ P_m^{\text{SF}} < p \tag{3.7}$$

over a range of the CPPI multiplier m values.[c]

In order to solve this problem numerically, we perform the following steps:

(1) Simulate 5000 paths of a risky asset.
(2) Simulate the CPPI strategy on each risky asset path with a chosen preliminary range of values of the multiplier m.
(3) For each value of m, approximate the shortfall probability in (3.1) by the frequency of the gap risk occurrences on the simulated paths.
(4) Determine the admissible range of the multiplier values based on the constraint (3.7).
(5) For each admissible value m apply the criterion (3.6) by approximating the expected values occurring in (3.4) with the appropriate averages over the simulated paths.
(6) Identify an optimal multiplier value m^*.

In this paper, we fix the guarantee level of the CPPI strategy at 90%: $P = 0.9$. This choice is determined by low interest rates of the current financial markets where guarantee rates higher than 90% are unrealistic from practitioners' perspective. However, the guarantee levels around 90% are considered reasonable since they provide for a sufficient budget to participate in the equity market.

We run the CPPI strategy based on 1-year contracts with 252 trading days. The portfolio is rebalanced daily regardless of the market movements[d] and assumes no transaction costs.

We use 2% as a risk-free rate of return in our calculations, unless otherwise is specified. This is a commonly used interest rate across the academic literature for numerical analysis and is extremely close in value to the average effective Federal funds rate in the past 30 years.

We limit the shortfall probability to 1%: $p = 0.01$.

In our simulations, we use the following range of values of the omega ratio threshold $L : 0\%, 1\%, 2\%, 3\%, 4\%$, and 5%.

[c]We use the slightly different notation here $\Omega_{X,m}(L)$ and P_m^{SF} to emphasize that both the omega ratio and the shortfall probability depend on the multiplier m.

[d]We limit our analysis to the daily closing prices of the S&P500 Index. However, it is worth mentioning that it is possible for the market to experience additional fluctuations in the Index price during the day. Such price movements are not captured by the day-to-day price observations but they, in practice, may trigger a portfolio manager to sell or buy equity before the scheduled rebalancing time.

The preliminary range of the multiplier values is as follows: integers from 2 to 22.

The approaches to the risky asset path simulation will be specified in the following two subsections.

3.4. *Discrete time trading with the Black–Scholes assumptions*

In this section, we illustrate our optimization algorithm for the case of a risky asset that follows the Black–Scholes assumptions [see (2.6)]. We simulate risky asset paths with various combinations of input parameters that reflect corresponding market scenarios. We implement our optimization algorithm assuming discrete time rebalancing and compare our numerical results with the theoretical formulas (see Sec. 2.2).

For our numerical simulations, we chose four combinations of the drift μ and annual volatility σ of a risky asset that are presented in Table 1. These combinations of parameters correspond to four different financial market scenarios. For each scenario, we simulate 5000 risky asset paths.

Scenarios 1 and 2 represent an equity market with low expected returns and with low and high volatility levels, respectively. Scenarios 3 and 4 portrait more optimistic market conditions with high expected returns combined with low and high volatility levels, respectively. We follow Dichtl & Drobetz (2010) in the choice of the risk premium of 4.5% for the markets with low expected returns (Scenarios 1 and 2) and the risk premium of 7% for the markets with high expected returns (Scenarios 3 and 4). We then calculate the expected annual risky asset return μ as a sum of the equity risk premium and the risk-free rate $r = 2\%$. Therefore, the expected return is $\mu = 6.5\%$ for the first two scenarios and is $\mu = 9\%$ for the last two scenarios. The low volatility market is modeled with $\sigma = 20\%$ and the high volatility market is modeled with $\sigma = 35\%$.

In Table 2, we report the shortfall probability for all four simulated scenarios. For comparison, the theoretical shortfall probability (denoted as $P^{\text{SF}}_{\text{theoretical}}$) calculated by means of (3.2) is also included. For each scenario, the estimated shortfall probability (denoted as $P^{\text{SF}}_{\text{simulated}}$) on simulated paths is very close to its theoretical counterpart: both simulated and theoretical values indicate the same cut-off point for a multiplier (in bold in Table 2). We only report results for the multiplier values

Table 1. The choice of the drift μ and volatility σ for the Black–Scholes dynamic of the underlying risky asset.

	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Drift	$\mu = 6.5\%$	$\mu = 6.5\%$	$\mu = 9\%$	$\mu = 9\%$
Volatility	$\sigma = 20\%$	$\sigma = 35\%$	$\sigma = 20\%$	$\sigma = 35\%$

Notes: Scenarios model four different equity market behaviors. Scenarios 1 and 2 have low expected returns combined with low and high volatility levels, respectively. Scenarios 3 and 4 have high expected returns combined with low and high volatility levels, respectively.

Table 2. The theoretical and calculated gap risk for four simulated scenarios.

	$m = 11$	$m = 12$	$m = 13$	$m = 14$	$m = 15$	$m = 16$
Scenario 1: $\mu = 6.5\%$ $\sigma = 20\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Scenario 2: $\mu = 6.5\%$ $\sigma = 35\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	0.002	**0.01**	0.035	0.094	0.200	0.353
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	0.005	**0.017**	0.059	0.139	0.259	0.446
Scenario 3: $\mu = 9\%$ $\sigma = 20\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	0.001
Scenario 4: $\mu = 9\%$ $\sigma = 35\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	0.002	**0.010**	0.035	0.093	0.197	0.349
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	0.004	**0.017**	0.055	0.135	0.264	0.439
	$m = 17$	$m = 18$	$m = 19$	$m = 20$	$m = 21$	$m = 22$
Scenario 1: $\mu = 6.5\%$ $\sigma = 20\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	< 0.001	0.001	0.002	0.006	**0.013**	0.027
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	< 0.001	0.001	0.003	0.009	**0.020**	0.039
Scenario 2: $\mu = 6.5\%$ $\sigma = 35\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	0.532	0.703	0.836	0.922	0.968	0.988
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	0.635	0.790	0.903	0.958	0.986	0.994
Scenario 3: $\mu = 9\%$ $\sigma = 20\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	< 0.001	0.001	0.002	0.005	**0.013**	0.026
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	0.001	0.001	0.003	0.009	**0.023**	0.038
Scenario 4: $\mu = 9\%$ $\sigma = 35\%$						
$P^{\mathrm{SF}}_{\mathrm{theoretical}}$	0.527	0.698	0.833	0.920	0.967	0.988
$P^{\mathrm{SF}}_{\mathrm{simulated}}$	0.631	0.790	0.904	0.957	0.984	0.995

Notes: $P^{\mathrm{SF}}_{\mathrm{theoretical}}$ is the theoretical shortfall probability calculated as in (3.2). $P^{\mathrm{SF}}_{\mathrm{simulated}}$ is the shortfall probability estimated as the frequency of the gap risk occurrences on the simulated paths. The numbers in bold indicate the shortfall probabilities that exceed the allowed gap risk threshold of 1%. For each scenario, the multipliers greater or equal to the multiplier that corresponds to the bold number are not included in the admissible range for further optimization.

$m = 10$ and higher since for the lower multiplier values the shortfall probability is below 0.001.

It is evident from the results presented in Table 2 that the admissible range of the multiplier values is highly sensitive to the risky asset volatility σ, but it is not significantly affected by the expected return μ of the risky asset. Indeed, for the cases with low volatility levels (Scenarios 1 and 3), an investor can choose a multiplier as high as $m = 20$ and still maintain the shortfall probability under 1%.

Table 3. Monte Carlo simulation results for multipliers $m = 2, \ldots, 11$: Scenarios 1, 2, 3, and 4.

	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$	$m=7$	$m=8$	$m=9$	$m=10$	$m=11$
Scenario 1: $\mu = 6.5\%$ $\sigma = 20\%$										
Return	3.33	4.05	4.82	5.65	6.57	7.61	8.81	10.28	12.11	14.44
$\Omega(0\%)$	**5.96**	4.01	3.31	2.96	2.77	2.66	2.59	2.56	2.55	2.56
$\Omega(1\%)$	**3.20**	2.65	2.42	2.30	2.24	2.20	2.19	2.20	2.22	2.24
$\Omega(2\%)$	1.83	1.82	1.82	1.83	1.84	1.86	1.88	1.91	1.95	1.99
$\Omega(3\%)$	1.10	1.29	1.40	1.48	1.54	1.59	1.63	1.68	1.73	1.78
$\Omega(4\%)$	0.68	0.94	1.10	1.21	1.30	1.37	1.44	1.50	1.55	1.61
$\Omega(5\%)$	0.44	0.70	0.88	1.01	1.11	1.20	1.27	1.34	1.40	1.46
Scenario 2: $\mu = 6.5\%$ $\sigma = 35\%$										
Return	3.27	3.86	4.39	4.80	5.04	5.03	4.69	3.88	2.50	0.56
$\Omega(0\%)$	**2.78**	2.27	2.10	2.07	2.14	2.27	2.42	2.52	2.51	2.32
$\Omega(1\%)$	1.95	1.78	1.74	1.78	1.87	2.01	2.16	**2.27**	2.26	2.10
$\Omega(2\%)$	1.42	1.43	1.47	1.55	1.66	1.80	1.95	**2.06**	**2.06**	1.92
$\Omega(3\%)$	1.06	1.17	1.26	1.36	1.48	1.63	1.78	1.88	**1.89**	1.76
$\Omega(4\%)$	0.80	0.97	1.09	1.21	1.34	1.49	1.63	1.73	**1.75**	1.63
$\Omega(5\%)$	0.62	0.82	0.96	1.08	1.22	1.36	1.50	1.61	**1.62**	1.52
Scenario 3: $\mu = 9\%$ $\sigma = 20\%$										
Return	3.90	4.95	6.09	7.34	8.75	10.36	12.28	14.60	17.47	21.03
$\Omega(0\%)$	**8.46**	5.71	4.76	4.32	4.08	3.94	3.87	3.83	3.80	3.78
$\Omega(1\%)$	**4.52**	3.78	3.49	3.36	3.30	3.27	3.27	3.28	3.30	3.31
$\Omega(2\%)$	2.59	2.61	2.64	2.67	2.71	2.76	2.81	2.86	2.90	2.94
$\Omega(3\%)$	1.57	1.86	2.04	2.17	2.27	2.36	2.45	2.52	2.58	2.63
$\Omega(4\%)$	0.98	1.36	1.60	1.79	1.93	2.05	2.15	2.24	2.31	2.38
$\Omega(5\%)$	0.64	1.01	1.29	1.49	1.66	1.79	1.91	2.01	2.09	2.16
Scenario 4: $\mu = 9\%$ $\sigma = 35\%$										
Return	3.77	4.67	5.50	6.17	6.57	6.64	5.43	3.96	1.84	9.80
$\Omega(0\%)$	**3.16**	2.57	2.34	2.23	2.13	2.01	1.84	1.60	1.30	0.96
$\Omega(1\%)$	**2.22**	2.02	1.95	1.91	1.86	1.78	1.64	1.43	1.17	0.87
$\Omega(2\%)$	1.61	1.63	1.65	**1.66**	1.65	1.59	1.48	1.30	1.06	0.79
$\Omega(3\%)$	1.21	1.34	1.42	1.46	**1.47**	1.44	1.34	1.18	0.97	0.72
$\Omega(4\%)$	0.92	1.11	1.23	1.30	**1.33**	1.30	1.23	1.09	0.89	0.67
$\Omega(5\%)$	0.72	0.94	1.08	1.16	**1.20**	1.19	1.13	1.00	0.83	0.62

Notes: *Return* is an average annual return of the investment with values reported as %. Ω is the omega ratio with a threshold specified in the parenthesis. The numbers in bold indicate the best values of omega performance measure within an admissible range of the multipliers. For each omega ratio threshold, the bold number corresponds to the optimal value of the multiplier.

However, when the volatility is 35% (Scenarios 2 and 4), the multiplier should not exceed $m = 11$ in order to keep the shortfall probability below 0.01.

Summarizing the results of the first stage of our optimization algorithm, the admissible ranges for the multiplier values are as follows: from $m = 2$ to $m = 20$ for Scenarios 1 and 3, and from $m = 2$ to $m = 11$ for Scenarios 2 and 4.

The results of the CPPI performance with multipliers $m = 2$ to $m = 11$ are summarized in Table 3. For the multipliers $m = 12$ to $m = 20$, only results for Scenarios 1 and 3 are reported (see Table 4) since high levels of volatility in Scenarios 2

Table 4. Monte Carlo simulation results for multipliers $m = 12, \ldots, 20$: Scenarios 1 and 3.

	$m = 12$	$m = 13$	$m = 14$	$m = 15$	$m = 16$	$m = 17$	$m = 18$	$m = 19$	$m = 20$
Scenario 1: $\mu = 6.5\%$ $\sigma = 20\%$									
Return	17.42	21.13	25.59	30.69	36.02	41.15	45.36	47.92	48.14
$\Omega(0\%)$	2.57	2.57	2.56	2.52	2.44	2.32	2.16	1.94	1.69
$\Omega(1\%)$	2.27	2.28	2.28	2.26	2.20	2.09	1.95	1.76	1.53
$\Omega(2\%)$	2.02	2.05	**2.06**	2.04	1.99	1.90	1.77	1.60	1.40
$\Omega(3\%)$	1.82	1.86	**1.87**	1.86	1.82	1.74	1.63	1.47	1.29
$\Omega(4\%)$	1.66	1.69	**1.71**	**1.71**	1.68	1.61	1.50	1.36	1.19
$\Omega(5\%)$	1.51	1.56	**1.58**	**1.58**	1.55	1.49	1.39	1.26	1.11
Scenario 3: $\mu = 9\%$ $\sigma = 20\%$									
Return	25.41	30.71	36.89	43.77	50.95	57.83	63.50	67.42	68.49
$\Omega(0\%)$	3.76	3.72	3.66	3.56	3.42	3.22	2.98	2.70	2.37
$\Omega(1\%)$	3.32	3.30	3.26	3.18	3.06	2.90	2.69	2.43	2.14
$\Omega(2\%)$	**2.96**	**2.96**	2.94	2.88	2.78	2.63	2.44	2.22	1.95
$\Omega(3\%)$	2.67	**2.68**	2.67	2.62	2.53	2.41	2.24	2.03	1.79
$\Omega(4\%)$	2.42	**2.44**	**2.44**	2.40	2.33	2.22	2.07	1.88	1.66
$\Omega(5\%)$	2.21	**2.24**	**2.24**	2.22	2.15	2.05	1.92	1.74	1.54

Notes: *Return* is an average annual return of the investment with values reported as %. Ω is the omega ratio with a threshold specified in the parenthesis. The numbers in bold indicate the best values of omega performance measure within an admissible range of the multipliers. For each omega ratio threshold, the bold number corresponds to the optimal value of the multiplier.

and 4 lead to an intolerable risk of breaching the capital protection guarantee when a multiplier is above $m = 11$.

For mildly volatile markets ($\sigma = 20\%$), the higher multiplier leads to a higher average CPPI return. The strategy with the lowest multiplier $m^* = 2$ on average scores the highest for omega ratios with low thresholds $L = 0\%$ and $L = 1\%$. Higher multipliers $m^* = 13$ and $m^* = 14$ are optimal for higher omega ratio thresholds $L = 2\%, 3\%, 4\%$, or 5%. This suggests that investors that are averse to any returns lower than the risk-free rate $r = 2\%$ should rely on higher multipliers.

In highly volatile markets ($\sigma = 35\%$), the CPPI portfolio with the multiplier $m = 6$ has the highest average yearly returns. Similar to the case of a mildly volatile market, the multiplier $m^* = 2$ is optimal for low omega ratio threshold $L = 0\%$. When the threshold rises, the preference of the CPPI strategy depends on the average market returns: in the case when $\mu = 6.5\%$, the strategies with multipliers $m^* = 9$ and $m^* = 10$ are optimal. While in a more optimistic market environment ($\mu = 9\%$), the multiplier $m^* = 6$ is optimal.

3.5. *Bootstrap simulation*

In addition to the Black–Scholes environment, we show how our optimization method works for the simulations in which the paths of the risky asset are bootstrapped in one-year blocks from the historical daily stock prices. This method allows us to estimate the optimal multiplier for the CPPI strategy for future investments without a need to impose any particular assumptions on the stock market return distribution.

While the empirical works are often based on the geometric Brownian motion simulations, it is a well-known fact that on the actual financial market stock returns do not follow the Black–Scholes assumptions. The distributions of stock returns are often negatively skewed with fat tails and therefore are not normal (Kon 1984). The daily observations of stock prices show autocorrelation and mean-reversion (Poterba & Summers 1988), and heteroscedasticity (Akgiray 2007). This invalidates the conventional statistical analysis based on the normality assumption.

The bootstrap simulations become widely popular technique in financial research. For example, Ledoit & Wolf (2008) show the benefits of the block-bootstrap simulations when one needs to apply a statistical test comparing two investments with regard to some performance measure such as Sharpe ratio. The method works well with time series data allowing to preserve the structure of the stock market returns: fat tails, skewness, autocorrelation, and volatility clustering. It recently finds successful applications in the numerical studies of the portfolio insurance strategies including the CPPI strategy (see Annaert *et al.* 2009, Bertrand & Prigent 2011, Zhang *et al.* 2015, Dichtl *et al.* 2017).

Our analysis is based on the daily historical data on S&P 500 tracker fund (100% correlated with S&P500 Index) from January 3rd, 2000 to June 10th, 2016. This timeframe covers several World economic crises, particularly the dot-com bubble in 2000–2002 and the global financial crisis of 2007–2008. The summary of the S&P 500 returns characteristics is presented in Table 5. The skewness and kurtosis of the return distribution confirm that the market stock prices are negatively skewed albeit mildly and observe fat tails. Jarque–Bera test is statistically significant meaning that the distribution fails the normality test (see Table 6). Engles Lagrange multiplier test indicates that there is heteroscedasticity and Ljung–Box test confirms

Table 5. Summary statistics of the S&P500 returns over the time period from January 3rd, 2000 to June 10th, 2016.

	Average annualized return	Annualized volatility	Minimum	Skewness	Kurtosis
S&P500	4.14%	19.87%	−9.03%	−0.005	8.33

Notes: Average annualized return is calculated as the geometric average return per year. Annualized volatility is calculated as the standard deviation of the daily returns multiplied by $\sqrt{252}$.

Table 6. Analysis of the S&P500 return distribution.

	p-value for normality test (Jarque–Bera test)	Autocorrelation at 1/2/3 lags (Ljung–Box test)	p-value for heteroscedasticity (Engle's Lagrange multiplier test)
S&P500	0.000	−0.08***/−0.04***/0.04***	0.000

Notes: ***Statistical significance at the 1% level.

autocorrelation in the data. This analysis provides evidence that Monte Carlo simulations based on the geometric Brownian motion of the risky asset assume unrealistic setup for the financial market environment and that a method that assumes no particular return distribution is preferred.

Following the methodology described in Annaert *et al.* (2009), Bertrand & Prigent (2011), and Dichtl *et al.* (2017), we use historical data (the total of 3930 one year blocks of Index prices) and run 5000 block-bootstraps with replacement, obtaining a sample of 5000 paths. It is a sufficient number of simulations for most of the numerical applications (Ledoit & Wolf 2008). Our block-bootstrap method implies that a starting date is randomly drawn with replacement from the sample and a path is constructed by taking 252 consecutive return observations from that date. As a result, we obtain a new sample of 5000 years. We compute the terminal CPPI values for each of the 5000 paths and use the distributions of the CPPI portfolio returns to compare the CPPI performance depending on the parameter m. As an interest rate for computing the floor during the CPPI rebalancing, we use the historical effective Federal Fund rate as of the day of rebalancing.

For strategies with multipliers $m = 12$ and higher, the shortfall probability (denoted as $P^{\mathrm{SF}}_{\mathrm{bootstrap}}$) is 0.08 and above, which is higher than the threshold $p = 0.01$ (see Table 7). Therefore, only the CPPI strategies with multipliers between $m = 2$ and $m = 11$ are considered for further optimization.

The performance of the CPPI strategy on the bootstrapped paths is summarized in Table 8. On average, the CPPI portfolios with the multiplier $m = 10$ produce the highest return of 5.02%, compared to the lowest return of 2.52% for $m = 2$. Consistent with the results of the simulations based on the Black–Scholes assumptions, omega measures with the low thresholds $L = 0\%$ and $L = 1\%$ indicate the preference for the lowest multiplier $m^* = 2$ and omega measures with thresholds at the level of the interest rate and greater ($L \geq 2\%$) are the highest in cases when multipliers are $m^* = 9$ and $m^* = 10$.

The results of our bootstrap simulation are consistent with the Black–Scholes simulations with high volatility levels (Scenarios 2 and 4) discussed in Sec. 3.3. Both approaches lead to the same upper limit of $m = 11$ for the admissible set of multipliers. Moreover, when in addition to high volatility the market produces rather low average returns (Scenario 2 in Sec. 3.4), the multipliers $m^* = 9$ and

Table 7. The gap risk in the CPPI portfolios based on bootstrap simulations.

	$m=8$	$m=9$	$m=10$	$m=11$	$m=12$	$m=13$	$m=14$	$m=15$
$P^{\mathrm{SF}}_{\mathrm{bootstrap}}$	<0.001	<0.001	<0.001	<0.001	**0.07**	0.07	0.07	0.07

Notes: $P^{\mathrm{SF}}_{\mathrm{bootstrap}}$ is the shortfall probability estimated as the frequency of the gap risk occurrences on the bootstrapped paths. The numbers in bold indicate the shortfall probabilities that exceed the allowed gap risk threshold of 1%. For each scenario, the multipliers greater or equal to the multiplier that corresponds to the bold numbers are not included in the admissible set for further optimization.

Table 8. Bootstrap simulation results.

	$m = 2$	$m = 3$	$m = 4$	$m = 5$	$m = 6$	$m = 7$	$m = 8$	$m = 9$	$m = 10$	$m = 11$	$m = 12$
Return	2.52	2.84	3.22	3.63	4.04	4.40	4.70	4.92	5.02	5.00	4.85
$\Omega(0\%)$	**5.32**	3.44	2.85	2.62	2.50	2.44	2.38	2.32	2.25	2.16	2.06
$\Omega(1\%)$	**2.78**	2.22	2.05	1.99	1.98	1.97	1.96	1.93	1.89	1.83	1.75
$\Omega(2\%)$	1.44	1.44	1.48	1.53	1.58	1.61	**1.62**	**1.62**	1.59	1.56	1.50
$\Omega(3\%)$	0.71	0.93	1.07	1.18	1.26	1.31	1.35	**1.36**	**1.36**	1.33	1.29
$\Omega(4\%)$	0.34	0.59	0.77	0.91	1.01	1.08	1.13	**1.16**	**1.16**	1.15	1.12
$\Omega(5\%)$	0.17	0.37	0.55	0.70	0.81	0.89	0.95	0.99	**1.00**	**1.00**	0.98

Notes: *Return* is an average annual return of the investment with values reported as %. Ω is the omega ratio with a threshold specified in the parenthesis. The numbers in bold indicate the best values of omega performance measure within an admissible range of the multipliers. For each omega ratio threshold, the bold number corresponds to the optimal value of the multiplier.

$m^* = 10$ are optimal for the omega ratio thresholds of at least 2%. These results agree with the results of the Monte Carlo simulations presented in Sec. 3.4.

4. Illustrative Numerical Example

In this section, we illustrate the results of our optimization algorithm applied to the real historical market data, daily returns of the S&P500 Index. We use two-year historical periods of the market S&P500 data. Each two-year period is split into two one-year periods. We use the first of the one-year periods as the training data in order to find an optimal multiplier for the CPPI strategy according to our optimization algorithm described in Sec. 3.3. We use the second one-year period as the testing period, where we apply the optimized CPPI strategy to the real market S&P500 data and evaluate the performance of the optimized CPPI strategy in comparison with the performance of all other admissible CPPI strategies.

The four two-year historical periods mentioned above are chosen in order to illustrate four different market trends: (1) a falling market, (2) first rising and then falling market, (3) first falling and then rising market, and (4) a rising market. The corresponding graphs are shown in Fig. 1. The summary for the chosen time periods is presented in Table 9.

At the first step of our optimization algorithm, we determine the upper bound for the admissible set of multipliers (see Sec. 3.3 for details). We simulate 1000 'future' S&P500 paths using stationary bootstrap from the training year and run the CPPI strategies with the multipliers from 2 to 22 on each of the simulated paths. Here, the effective Federal funds rate is used as a risk-free interest rate for the CPPI strategies simulations. The results of this first step of the optimization process are presented in the last column of Table 9.

Remark 4.1. At this step, we remove from our analysis the risky paths for which the CPPI returns are outliers. We consider an outlier any CPPI return greater than 1.5 times interquartile range added to the third quartile of the return distribution.

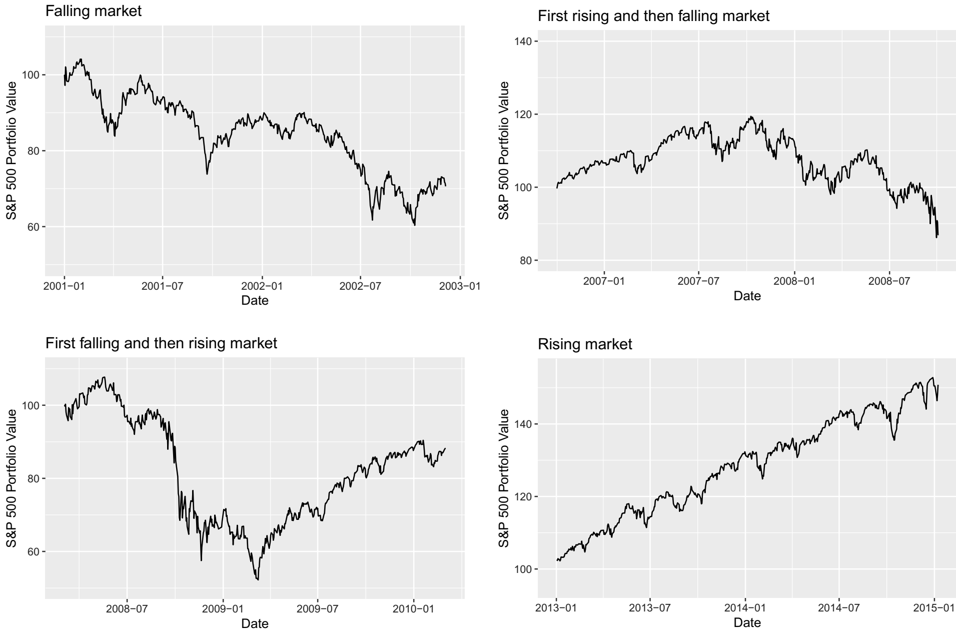

Fig. 1. Price trends of the S&P500 Index in four different two-year periods. The initial investment in the S&P500 portfolio at the beginning of each period is $100. The plots reflect daily S&P500 prices.

Table 9. Summary of the analyzed two-year periods.

Market pattern	Start of training year	Start of testing year	S&P500 return in testing year (%)	Interest rate in testing year (%)	Upper bound for multiplier
Falling	Jan 1, 2001	Dec 19, 2001	−18.45	1.71	17
Rising then falling	Oct 2, 2006	Oct 4, 2007	−23.29	4.74	17
Falling then rising	Mar 3, 2008	Mar 4, 2009	30.37	0.21	11
Rising	Jan 2, 2013	Jan 3, 2014	12.94	0.08	11

Notes: All training and testing years are 252-trading day long. The upper bound for the admissible set of multipliers is calculated by the optimization algorithm.

In identifying outliers, we focus on the CPPI returns when a multiplier $m = 2$. This results in removing the minimum necessary set of outliers.

The second step of the optimization algorithm identifies the optimal multiplier m^* depending on a threshold L in omega ratio (see Sec. 3.3 for details). We find optimal m^* for the thresholds $L = 0\%, 1\%, 2\%, \ldots, 10\%$.

After the completion of the optimization process, we run the CPPI strategies with all admissible multipliers (including the optimal multiplier) on the testing year

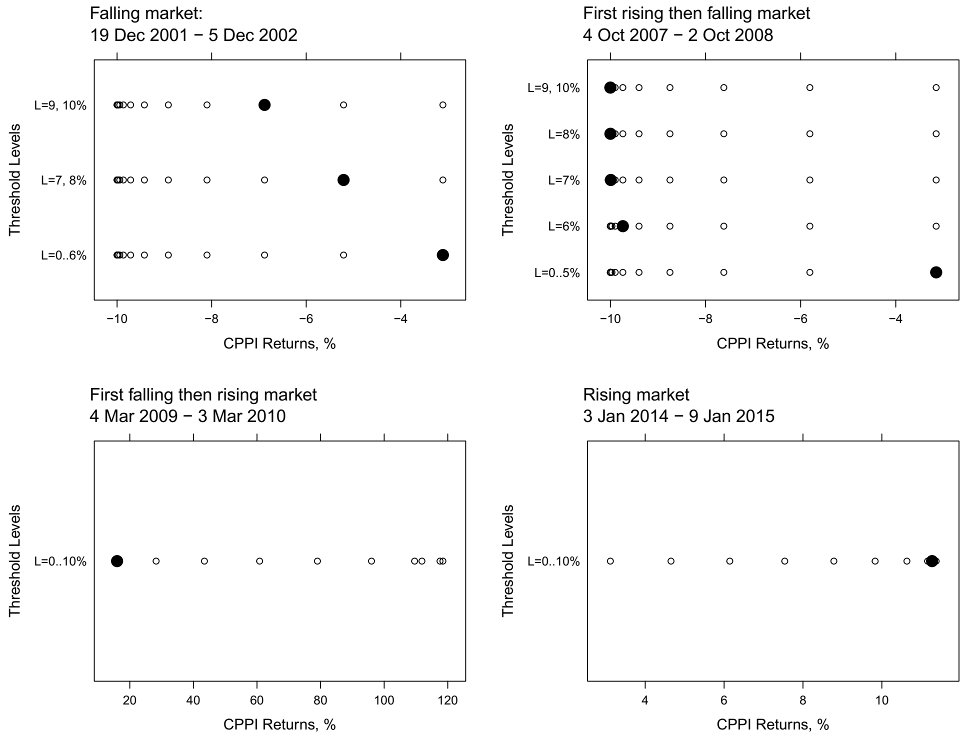

Fig. 2. Performance of the CPPI strategies with different multipliers in different real market scenarios. Each circle represents a CPPI return for a given multiplier. Solid black dots represent the CPPI returns with the optimized multipliers.

data (the second one-year period in each market pattern). As a result, we obtain one CPPI portfolio return per multiplier value for each threshold. Finally, we compare the return of the optimized CPPI portfolio with the returns of the CPPI portfolios with all the remaining admissible multipliers.

The performances of the CPPI strategies with different multipliers in different market scenarios are visualized in Fig. 2. Each plot corresponds to a particular market pattern. Each circle represents a CPPI portfolio return for a multiplier from the admissible range. A solid black dot corresponds to the CPPI portfolio with the optimized multiplier m^*. The number of strips in each plot corresponds to the number of different optimal multiplier values m^* within the range of the threshold values $L = 0\%, 1\%, 2\%, \ldots, 10\%$.

The results of these numerical experiments suggest that the optimization algorithm based on bootstrap simulations performs the best in scenarios with consistent market trends throwout both the training and testing periods. This may be observed in the plots for the consistently falling (the upper left plot) and consistently rising (the lower right plot) markets in Fig. 2. In the consistently rising market case, the optimized CPPI strategy produces the highest return among the competing CPPI strategies for all threshold levels. In the case of consistently falling market, the performance of the optimized CPPI strategy depends on the investor's choice of the omega threshold L. In the case when the value of L is chosen around the risk-free rate (a reasonable choice in this scenario), the optimized CPPI strategy yields the highest return. With increasing threshold levels, the return produced by the optimized CPPI strategy decreases, but the optimized CPPI strategy still performs among the best admissible strategies.

The optimized CPPI strategy also performs well in the case when the market first rises and then experiences a fall (the upper right plot), and the choice of the omega threshold L is around the risk-free interest rate. In this case, the optimized strategy produces the highest return. However, with increasing threshold level, the performance of the optimized CPPI portfolio significantly decreases.

In the case of the market which falls over the training period and then rises over the testing period (the lower left plot), the optimized CPPI strategy does not perform well. The optimization algorithm favors the most conservative multiplier $m^* = 2$, which prevents the CPPI portfolio from gaining when the market rises over the testing period.

5. Conclusion

In this paper, we have addressed the optimization problem for the multiplier of the CPPI strategy. We have proposed an optimization algorithm that allows to maintain the gap risk under a prespecified limit and identifies an optimal multiplier with respect to the omega ratio criterion within the admissible range. We first have imposed an upper bound on the multiplier range such that the shortfall probability of the CPPI strategy is limited by a small number (e.g. 0.01). We then have used omega ratio as a comprehensive performance measure to select an optimal value of the multiplier given various threshold levels that reflect an investor's preference.

The proposed method has been implemented in two simulation environments: (1) the continuous trading assumption has been lifted, and more realistic market scenarios have been modeled through the discrete Monte Carlo simulations based on the geometric Brownian motion; and (2) both continuous trading and the Black–Scholes assumptions have been lifted with the block-bootstrap simulations based on historical market data.

To the best of our knowledge, this is the first work that offers a comprehensive framework to select an optimal multiplier for the simple CPPI strategy. The first simulation environment allows us to model various market scenarios as well as compare our simulation results with the theoretical ones. The second simulation environment provides a more realistic look at performance of our optimization algorithm in the real financial market environment.

As our simulation results in Sec. 4 show, the optimization of the CPPI strategy often leads to improved performance compared to CPPI strategies with multipliers chosen arbitrarily. However, it is important to note that the optimized strategy has this advantage not in all financial market patterns.

We have chosen to make the emphasis on the numerical optimization in order to relax restrictive market assumptions required for the theoretical analysis. While closed-form relationship between the multiplier and omega ratio would certainly contribute to the discussion on the optimal multiplier, we leave this question for the future research.

The optimization of the CPPI strategy may bring a new perspective to the ongoing competition among different portfolio insurance strategies. The further development of this topic may therefore include an extensive comparison of the optimized CPPI strategy to its main rival OBPI strategy.

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